Evaluation Methods in NLP

CS 115B: Fundamentals of NLP

James Pustejovsky

Brandeis University

Spring 2025

Today's Lecture

1. Why Evaluation Matters
2. Accuracy and Its Limitations
3. The Confusion Matrix
4. Precision, Recall, and F-Measure
5. Multi-Class Evaluation
6. Threshold-Free Metrics
7. Train/Dev/Test Splits and Cross-Validation
8. Statistical Significance Testing
9. Inter-Annotator Agreement
10. Avoiding Harms in Classification
11. Summary

Why Evaluation Matters

Evaluation is fundamental to NLP research and engineering:

- How do we know if our system works?
- How do we compare different approaches?
- How do we know if improvements are real or due to chance?
- How do we ensure our system works fairly for all users?

Key principles:

- Always reserve a **held-out test set**
- Use it **only once** for final evaluation
- Using test data more than once $\Rightarrow$ overly optimistic estimates
- Use a *development set* for tuning hyperparameters

Accuracy: The Simplest Metric

Accuracy: Number of correct predictions divided by total instances

$$\text{Accuracy}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(y^{(i)} = \hat{y}^{(i)})$$

Intuitive and easy to understand—like grading an exam.

Accuracy: The Simplest Metric

Accuracy: Number of correct predictions divided by total instances

$$\text{Accuracy}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(y^{(i)} = \hat{y}^{(i)})$$

Intuitive and easy to understand—like grading an exam.

So why do we need other metrics?

Accuracy: The Simplest Metric

Accuracy: Number of correct predictions divided by total instances

$$\text{Accuracy}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(y^{(i)} = \hat{y}^{(i)})$$

Intuitive and easy to understand—like grading an exam.

So why do we need other metrics?

The main problem: Class imbalance

The Problem with Accuracy: Class Imbalance

Example 1: Rare Disease Detection
Detecting symptoms in electronic health records where the disease appears in only 1% of documents.

A classifier that always predicts NEGATIVE achieves **99% accuracy**!
But it's **completely useless**—it never detects the disease.

The Problem with Accuracy: Class Imbalance

Example 1: Rare Disease Detection
Detecting symptoms in electronic health records where the disease appears in only 1% of documents.

A classifier that always predicts NEGATIVE achieves **99% accuracy**!
But it's **completely useless**—it never detects the disease.

Example 2: The Delicious Pie Company
You want to find tweets about your pies. Out of 1,000,000 tweets:

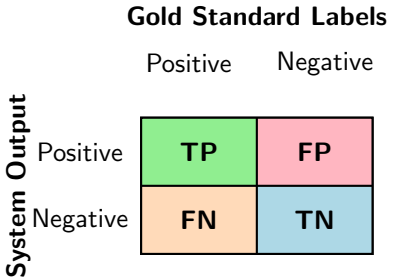
- 100 are about Delicious Pie
- 999,900 are about other things

A classifier that labels *every* tweet as "not about pie":

- 999,900 true negatives, 100 false negatives
- Accuracy: **99.99%** — but finds zero pie tweets!

The Confusion Matrix

A **confusion matrix** visualizes classifier performance against gold labels.



- **True Positive (TP)**: System correctly predicts label
- **False Positive (FP)**: System incorrectly predicts label
- **False Negative (FN)**: System fails to predict label
- **True Negative (TN)**: System correctly rejects label

Two Types of Errors

False Positives (Type I Error):

- System predicts the label when it isn't there
- "Crying wolf"
- Example: Flagging a legitimate email as spam

False Negatives (Type II Error):

- System fails to predict the label when it is there
- "Missing the mark"
- Example: Failing to detect a disease in a patient

Different applications have different costs for each error type!

Precision

Definition

Precision measures the percentage of items the system labeled as positive that are actually positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Intuition: "When the system says YES, how often is it right?"

High precision = Few false positives = System is **conservative**

When to prioritize precision:

- False positives are expensive
- Spam detection (don't want to lose important emails)
- Criminal justice ("innocent until proven guilty")

Recall

Definition

Recall (also called *sensitivity* or *true positive rate*) measures the percentage of actual positives that were correctly identified.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

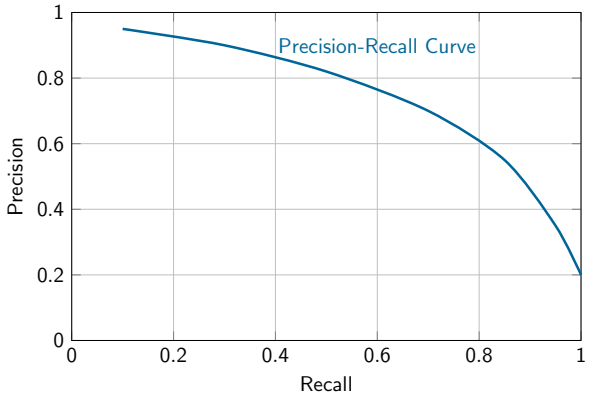
Intuition: "Of all the things that ARE positive, how many did we find?"

High recall = Few false negatives = System is **thorough**

When to prioritize recall:

- False negatives are expensive
- Disease screening (don't want to miss sick patients)
- Security threats (don't want to miss attacks)

The Precision-Recall Trade-off



Key insight: Precision and recall are often in tension!

- Being more aggressive ⇒ higher recall, lower precision
- Being more conservative ⇒ higher precision, lower recall

F-Measure: Combining Precision and Recall

Definition

The **F-measure** combines precision and recall using the **harmonic mean**:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2PR}{P + R}$$

Why harmonic mean?

- Harmonic mean is closer to the *minimum* of two values
- Penalizes extreme imbalances between P and R
- Arithmetic mean of $P = 0.9, R = 0.1$ is 0.5
- Harmonic mean (F1) is only 0.18!

General form: $F_\beta = \frac{(1+\beta^2)PR}{\beta^2 P + R}$

- $\beta > 1$: emphasize recall
- $\beta < 1$: emphasize precision
- $\beta = 1$: balance (most common)

Example: Computing Precision, Recall, F1

Scenario: Spam detection on 100 emails (20 actual spam, 80 not spam)

	Gold: Spam	**Gold: Not Spam**
Predicted: Spam	TP = 15	FP = 5
Predicted: Not Spam	FN = 5	TN = 75

$$\text{Precision} = \frac{15}{15+5} = \frac{15}{20} = 0.75$$

$$\text{Recall} = \frac{15}{15+5} = \frac{15}{20} = 0.75$$

$$F_1 = \frac{2 \times 0.75 \times 0.75}{0.75 + 0.75} = 0.75$$

$$\text{Accuracy} = \frac{15+75}{100} = 0.90$$

Multi-Class Classification

Many NLP tasks have **more than two classes**:

- Sentiment: positive, negative, neutral
- Email: urgent, normal, spam
- Named entities: person, organization, location, misc, O

How do we compute a single metric across multiple classes?

Two main approaches:

1. **Macro-averaging**: Average metrics across classes
2. **Micro-averaging**: Pool counts, then compute metrics

Multi-Class Confusion Matrix Example

3-way email classification: urgent, normal, spam

	Gold Labels			
System	urgent	normal	spam	
urgent	8	10	1	$P_u = \frac{8}{19} = .42$
normal	5	60	50	$P_n = \frac{60}{115} = .52$
spam	3	30	200	$P_s = \frac{200}{233} = .86$
	$R_u = \frac{8}{16} = .50$	$R_n = \frac{60}{100} = .60$	$R_s = \frac{200}{251} = .80$	

Macro-Averaging

Definition

Macro-averaging: Compute the metric for each class, then average.

$$\text{Macro-}F_1 = \frac{1}{K}\sum_{k=1}^{K} F_1(\mathbf{y}, \hat{\mathbf{y}}, k)$$

From our example:

$$\text{Macro-Precision} = \frac{.42 + .52 + .86}{3} = \mathbf{.60}$$

$$\text{Macro-Recall} = \frac{.50 + .60 + .80}{3} = \mathbf{.63}$$

Properties:

- Each class weighted **equally**
- Better reflects performance on **smaller classes**
- Use when all classes are equally important

Micro-Averaging

Definition

Micro-averaging: Pool TP, FP, FN across all classes, then compute metrics.

From our example:
Total TP $= 8 + 60 + 200 = 268$
Total FP $= (10 + 1) + (5 + 50) + (3 + 30) = 99$

$$\text{Micro-Precision} = \frac{268}{268 + 99} = \mathbf{.73}$$

Properties:

- Each *instance* weighted equally
- Dominated by **more frequent classes**
- Use when you care about overall instance-level performance

Macro vs. Micro: When to Use Which?

Macro-Averaging	**Micro-Averaging**
Weights classes equally	Weights instances equally
Sensitive to rare classes	Dominated by common classes
Use when all classes matter equally	Use when overall accuracy matters
Lower scores if rare classes are hard	May mask poor performance on rare classes

Example: In our email task, micro-precision (.73) is higher than macro-precision (.60) because we do well on the large "spam" class.

Always report both when class distributions are imbalanced!

ROC Curves

In binary classification, we can trade off precision and recall by adjusting a **threshold** on the classifier's score.

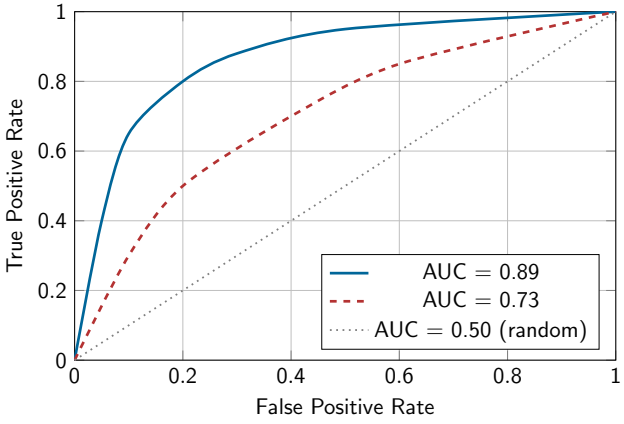
ROC Curve (Receiver Operating Characteristic):

- X-axis: **False Positive Rate** $= \frac{\text{FP}}{\text{FP}+\text{TN}}$
- Y-axis: **True Positive Rate** (Recall) $= \frac{\text{TP}}{\text{TP}+\text{FN}}$

Interpretation:

- **Perfect classifier**: Curve goes $(0,0) \to (0,1) \to (1,1)$
- **Random classifier**: Diagonal from $(0,0)$ to $(1,1)$
- **Real classifiers**: Between these extremes

ROC Curves: Visualization



Area Under the Curve (AUC)

Definition

AUC = Area Under the ROC Curve

Interpretation: Probability that a randomly-selected positive example scores higher than a randomly-selected negative example.

- **AUC = 1.0**: Perfect classifier
- **AUC = 0.5**: Random classifier (no discrimination)
- **AUC = 0.0**: Perfectly wrong classifier

Advantages of AUC:

- Threshold-free (summarizes all possible thresholds)
- Baseline of 0.5 **doesn't depend on class distribution**
- Good for comparing classifiers across different operating points

Data Splits

Standard practice: Split data into three sets



- **Training set**: Used to train the model
- **Development set** (dev/validation): Tune hyperparameters, feature selection
- **Test set**: Final evaluation **(use only once!)**

Why separate dev and test?

- Prevents "overfitting to the test set"
- Gives unbiased estimate of real-world performance

Cross-Validation

Problem: Fixed splits may waste data or be unrepresentative.

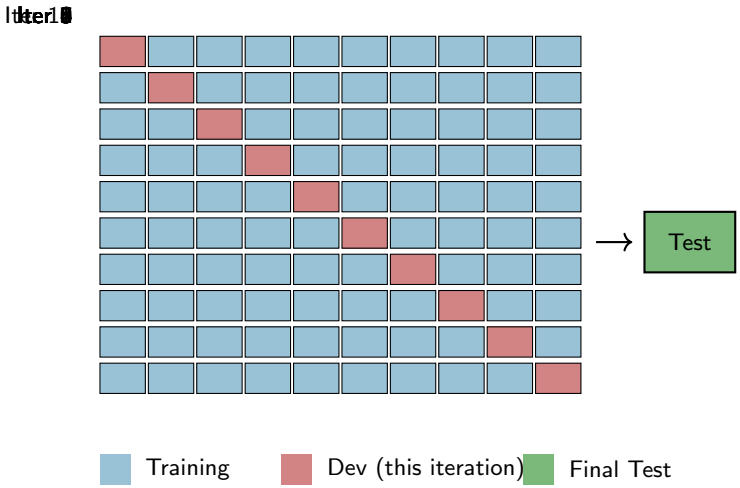
Solution: ***k*-fold cross-validation**

1. Partition data into k disjoint *folds*
2. For each fold i:
   - Use fold i as test set
   - Train on remaining $k - 1$ folds
   - Compute error on fold i
3. Average error across all k folds

Common choice: $k = 10$ (10-fold cross-validation)

- Train 10 models, each on 90% of data
- Every instance is tested exactly once

10-Fold Cross-Validation



Why Statistical Significance?

Scenario: Classifier A has F1 = 0.85, Classifier B has F1 = 0.81

Question: Is A really better, or did it just get lucky on this test set?

Why Statistical Significance?

Scenario: Classifier A has F1 = 0.85, Classifier B has F1 = 0.81

Question: Is A really better, or did it just get lucky on this test set?

The problem:

- A might be "accidentally" better on this particular test set
- On a different test set, B might do better
- We need to quantify our **confidence** in the difference

Why Statistical Significance?

Scenario: Classifier A has F1 = 0.85, Classifier B has F1 = 0.81

Question: Is A really better, or did it just get lucky on this test set?

The problem:

- A might be "accidentally" better on this particular test set
- On a different test set, B might do better
- We need to quantify our **confidence** in the difference

Statistical hypothesis testing helps us answer:
"How likely is it that the observed difference arose by chance?"

Hypothesis Testing Framework

Define the effect size:

$$\delta(x) = M(A, x) - M(B, x)$$

where M is our metric (e.g., F1) and x is the test set.

Two hypotheses:

- H_0 (**null hypothesis**): $\delta(x) \leq 0$ (A is not better than B)
- H_1 (alternative hypothesis): $\delta(x) > 0$ (A is better)

Goal: Determine if we can **reject** H_0.

***p*-value:** Probability of seeing $\delta(x)$ or larger, *assuming* H_0 *is true.*

$$p\text{-value} = P(\delta(X) \geq \delta(x) \mid H_0 \text{ is true})$$

Interpreting p-values

Common thresholds:

- $p < 0.05$: "Statistically significant"
- $p < 0.01$: "Highly significant"
- $p < 0.001$: "Very highly significant"

What does $p < 0.05$ mean?
If H_0 were true (A is not better than B), there's less than 5% chance we'd see a difference as large as we observed.

Interpreting p-values

Common thresholds:

- $p < 0.05$: "Statistically significant"
- $p < 0.01$: "Highly significant"
- $p < 0.001$: "Very highly significant"

What does $p < 0.05$ mean?
If H_0 were true (A is not better than B), there's less than 5% chance we'd see a difference as large as we observed.

Caution: p-values are often misinterpreted!

- $p = 0.05$ does NOT mean "5% chance H_0 is true"
- Statistical significance $\neq$ practical significance
- A large effect might not be significant (small test set)
- A tiny effect might be significant (huge test set)

The Binomial Test

For comparing accuracy of two classifiers:

Let N = number of instances where classifiers disagree
Let k = number of those where classifier c_1 is correct

Under H_0 (classifiers equally good): $k \sim \text{Binomial}(N, 0.5)$

$$p_{\text{Binom}}(k; N, \theta) = \binom{N}{k} \theta^k (1-\theta)^{N-k}$$

***p*-value (one-tailed):**

$$P(K \leq k \mid H_0) = \sum_{i=0}^{k} p_{\text{Binom}}(i; N, \theta = 0.5)$$

Two-tailed: Double the one-tailed p-value

The Paired Bootstrap Test

For arbitrary metrics (F1, BLEU, etc.) where binomial test doesn't apply.

Idea: Create many "virtual test sets" by resampling.

Algorithm:

1. Compute $\delta(x)$ on original test set x
2. For $i = 1$ to b (e.g., $b = 10^4$):
   - Create bootstrap sample $x^{(i)}$ by sampling n instances *with replacement*
   - Compute $\delta(x^{(i)})$
3. Count how often $\delta(x^{(i)}) \geq 2\delta(x)$
4. p-value $\approx \frac{1}{b} \sum_{i=1}^{b} \mathbf{1}[\delta(x^{(i)}) \geq 2\delta(x)]$

Why $2\delta(x)$? Because bootstrap samples are biased toward original $\delta(x)$.

Bootstrap Test: Example

	1	2	3	4	5	6	7	8	9	10	A%	B%	δ
x	AB	A$\not$B	$\not$AB	AB	A$\not$B	$\not$A$\not$B	AB	A$\not$B	AB	$\not$AB	.70	.50	.20
$x^{(1)}$	A$\not$B	AB	A$\not$B	AB	AB	A$\not$B	AB	AB	A$\not$B	AB	.60	.60	.00
$x^{(2)}$	A$\not$B	AB	A$\not$B	AB	AB	AB	AB	A$\not$B	AB	AB	.60	.70	-.10
$\vdots$													

Legend:

- AB = Both correct
- A$\not$B = A correct, B wrong
- $\not$AB = A wrong, B correct
- $\not$A$\not$B = Both wrong

If in only 47 of 10,000 samples $\delta(x^{(i)}) \geq 2\delta(x) = 0.40$:
p-value $= 0.0047 < 0.01 \Rightarrow$ **Reject** H_0, A is significantly better!

Multiple Comparisons Problem

Problem: Testing multiple hypotheses inflates false positive rate.

Example: Compare 4 classifiers on 5 datasets = 20 tests
At $\alpha = 0.05$: expect 1 false positive even if no real differences!

Solutions:

1. **Bonferroni Correction:**
   - Use threshold $\frac{\alpha}{m}$ for m tests
   - Very conservative
2. **False Discovery Rate (FDR):**
   - Benjamini-Hochberg procedure
   - Less conservative, bounds fraction of false discoveries
   - Sort p-values, find largest k where $p_k \leq \frac{k}{m}\alpha$

Building Datasets: Annotation

Many NLP tasks require human-annotated data.

Good annotation protocols should be:

- **Expressive**: Capture the phenomenon of interest
- **Replicable**: Another annotator would produce similar results
- **Scalable**: Can be produced relatively quickly

Annotation workflow:

1. Determine what to annotate (based on theory)
2. Select/design annotation tool (BRAT, Label Studio, etc.)
3. Write explicit instructions
4. Pilot annotation with multiple annotators
5. Main annotation effort
6. Compute and report inter-annotator agreement

Cohen's Kappa

Raw agreement is misleading—annotators might agree by chance!

Cohen's Kappa

$$\kappa = \frac{\text{observed agreement} - \text{expected agreement}}{1 - \text{expected agreement}}$$

Interpretation:

- $\kappa = 1$: Perfect agreement
- $\kappa = 0$: Agreement no better than chance
- $\kappa < 0$: Agreement worse than chance

Expected agreement (for 2 annotators):

$$E[\text{agreement}] = \sum_k P(\hat{Y} = k)^2$$

Interpreting Kappa

Example: Binary labeling, one label applied to 90% of instances
Expected agreement $= 0.9^2 + 0.1^2 = 0.82$
So 90% raw agreement is only $\kappa = \frac{0.90-0.82}{1-0.82} = 0.44$!

Rough guidelines (Landis & Koch, 1977):

κ	Interpretation
< 0	Poor
$0.0 - 0.20$	Slight
$0.21 - 0.40$	Fair
$0.41 - 0.60$	Moderate
$0.61 - 0.80$	Substantial
$0.81 - 1.00$	Almost perfect

Real examples:

- Emotion in speech: $\kappa \approx 0.45$–0.47 (moderate)
- Dialogue acts: $\kappa = 0.80$ (substantial)

Harms from NLP Classifiers

Classifiers can cause real-world harm:

Representational harms:

- Perpetuating negative stereotypes
- Example: Sentiment systems assign lower scores to sentences with African American names (Kiritchenko & Mohammad, 2018)

Allocation harms:

- Unfair distribution of resources or opportunities
- Example: Résumé screening systems biased against certain groups

Silencing:

- Toxicity classifiers incorrectly flag non-toxic content
- Higher false positive rates for African American Vernacular English
- Can suppress legitimate speech from marginalized groups

Sources of Bias

Bias can enter at many stages:

- **Training data**: Historical biases, unrepresentative samples
- **Labels**: Annotator biases, ambiguous guidelines
- **Resources**: Biased lexicons, pretrained embeddings
- **Model architecture**: What the model optimizes for
- **Evaluation**: Metrics that don't capture fairness

ML systems replicate and amplify biases in training data!

No general solutions yet—active area of research.

Model Cards

Model cards (Mitchell et al., 2019): Document ML models with:

- Training algorithms and parameters
- Training data sources, motivation, preprocessing
- Evaluation data sources, motivation, preprocessing
- **Intended use and users**
- **Model performance across demographic groups**
- Known limitations and failure modes
- Ethical considerations

Goal: Transparency about what models can and cannot do, and for whom.

Good practice: Release a model card with every model version!

Summary: Key Evaluation Concepts

Metrics:

- Accuracy is insufficient for imbalanced classes
- Precision, Recall, F1 capture different aspects of performance
- Macro-averaging weights classes equally; micro-averaging weights instances
- ROC/AUC provide threshold-free evaluation

Methodology:

- Train/Dev/Test splits prevent overfitting
- Cross-validation maximizes data usage
- Statistical significance tests validate improvements
- Inter-annotator agreement measures annotation quality

Ethics:

- Evaluate across demographic groups
- Document limitations with model cards

Evaluation Checklist

- ☐ Reserved held-out test set (used only once)
- ☐ Appropriate metrics for the task and class distribution
- ☐ Both precision and recall (or F1) reported
- ☐ Macro and micro averages for multi-class
- ☐ Statistical significance tests for comparisons
- ☐ Confidence intervals where appropriate
- ☐ Inter-annotator agreement for human-labeled data
- ☐ Performance broken down by relevant subgroups
- ☐ Model card documenting limitations and intended use

Questions?

James Pustejovsky
jamesp@brandeis.edu

References:

Eisenstein, *NLP Notes*, Chapter 4
Jurafsky & Martin, *Speech and Language Processing*, Chapter 4